



Barker College

Student Number:

**2010
YEAR 12
TRIAL HSC
EXAMINATION**

MATHEMATICS

Staff Involved:

THURSDAY 5TH AUGUST

- TRW • LJP
- GIC • JGD
- VAB • KJL
- GPF* • AJD
- WMD*

130 copies

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using blue or black pen
- Write your Barker Student Number on all pages of your answers
- Board-approved calculators may be used
- A Table of Standard Integrals is provided at the back of this paper which may be detached for your use
- ALL necessary working MUST be shown in every question
- Marks may be deducted for careless or badly arranged working

Total marks - 120

- Attempt Questions 1 - 10
- All questions are of equal value
- **BEGIN** your answer to EACH QUESTION on a NEW PIECE of the separate lined paper
- Write only on ONE SIDE of the separate lined paper

-BLANK PAGE -

Total marks - 120
Attempt Questions 1 - 10
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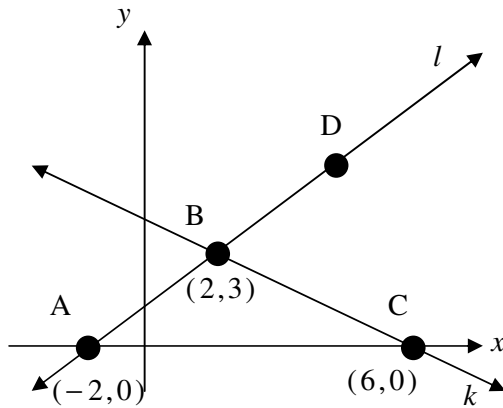
Answer each question on a separate A4 lined sheet of paper.

	Marks
Question 1 (12 marks) [START A NEW PAGE]	
(a) Evaluate, to 3 significant figures, $\frac{(-2.4)^2}{\sqrt{2\pi-5}}$.	2
(b) Simplify fully $\frac{16a^3-54b^3}{4a^2+6ab+9b^2}$.	2
(c) Solve for x : $ 2x-5 < 8$.	2
(d) Write down the domain of the function $y = \frac{1}{\sqrt{9-x}}$.	1
(e) Solve for x : $9^x - 7(3^x) - 18 = 0$.	3
(f) Determine whether $f(x) = \frac{x}{x^2-3}$ is odd, even or neither, justifying your answer with appropriate working.	2

End of Question 1

Question 2 (12 marks) [START A NEW PAGE]

The diagram shows the lines l and k :



NOT TO SCALE.

Redraw this diagram in your answer booklet.

- | | | |
|-------|--|---|
| (i) | Calculate the gradient of the line l . | 1 |
| (ii) | Calculate the length of AB . | 1 |
| (iii) | Find in general form , the equation of the line l . | 2 |
| (iv) | Calculate the angle of inclination of the line l (to the nearest degree). | 1 |
| (v) | D is a point on the line l , such that AD and CD are perpendicular.
Find the coordinates of D . | 4 |
| (vi) | Show the area of the $\triangle ABC$ is 12 square units. | 1 |
| (vii) | Find $\angle ABC$ (to the nearest degree). | 2 |

End of Question 2

Question 3 (12 marks) **[START A NEW PAGE]**

(a) Differentiate the following expressions with respect to x :

(i) $\sqrt{x} \ln x$ 2

(ii) $\tan\left(\frac{\pi}{2} - x^2\right)$ 2

(b) Find $\frac{d}{dx}\left[\frac{x}{e^{x^2}}\right]$ in simplest form. 3

(c) Find the values of k for which $x^2 + (k + 3)x - k = 0$ has real roots. 3

(d) If α and β are the roots of $2x^2 - 5x + 1 = 0$,
find the value of $\alpha^2 + \beta^2$. 2

End of Question 3

Question 4 (12 marks) **[START A NEW PAGE]**

- (a) For the function with the equation $y = 3\sin\left(\frac{x}{2} + \frac{\pi}{4}\right)$
- (i) State the amplitude of the function. **1**
 - (ii) State the period of the function. **1**
 - (iii) Sketch the graph of the function over the domain $0 \leq x \leq 2\pi$. **2**
- (b) Solve the equation $4\sin x = 3\operatorname{cosec} x$ for $0^\circ \leq x \leq 360^\circ$. **3**
- (c) Prove $\frac{\cos \alpha}{1 + \sin \alpha} = \sec \alpha (1 - \sin \alpha)$. **3**
- (d) Find the 32nd term of the series $(-6) + (-2) + 2 + \dots$ **1**
- (e) Find the limiting sum of the series $108 + 36 + 12 + \dots$ **1**

End of Question 4

Question 5 (12 marks) [START A NEW PAGE]

(a) Find $\int \frac{dx}{\sqrt[3]{x^2}}$. 2

(b) Evaluate $\int_2^{\sqrt{7}} \frac{x}{x^2-3} dx$. 3

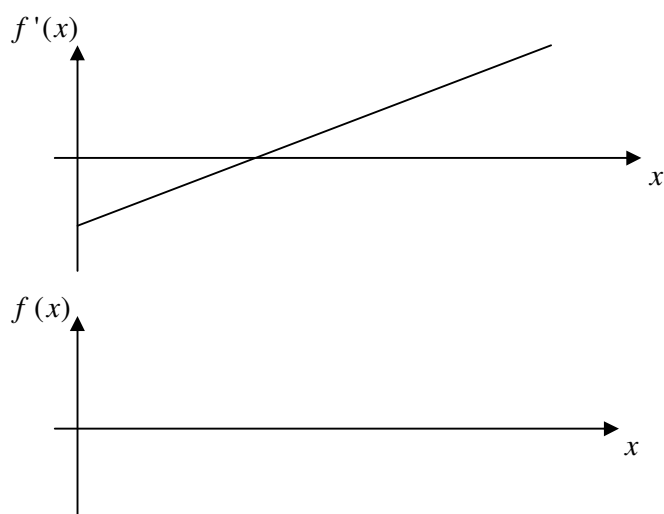
- (c) (i) **In your answer booklet**, copy and complete the table below for the function $f(x) = e^x$, giving your answers to 3 decimal places where necessary. 1

x	0	1	2
$f(x)$			

- (ii) Using Simpson's rule with 3 function values, find an approximation to $\int_0^2 e^x dx$. 2

- (d) Find the volume of the solid of revolution formed by rotating the line $y = -2x$ between $x = 1$ and $x = 5$ about the x -axis. 3

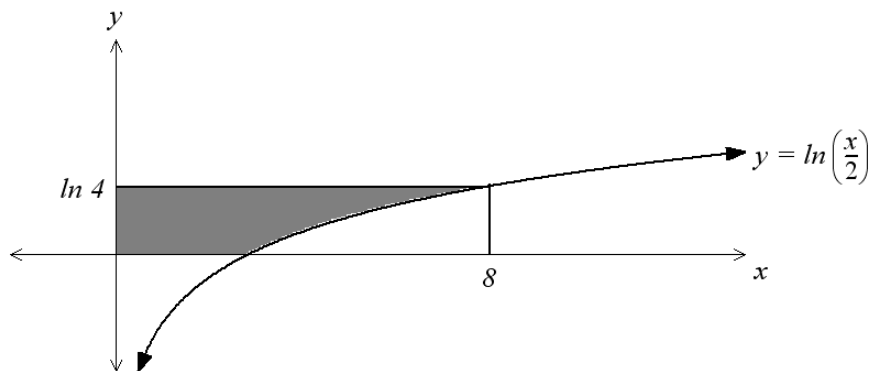
- (e) **In your answer booklet**, copy the diagrams below. Use the graph of $y = f'(x)$ to complete a possible graph of $y = f(x)$. 1



End of Question 5

Question 6 (12 marks) **[START A NEW PAGE]**

(a)



The diagram shows the area bounded by the graph $y = \ln\left(\frac{x}{2}\right)$, the co-ordinates axes and the line $y = \ln 4$.

Find the area of the shaded region.

3

(b) The curve $y = f(x)$ has a gradient function of $\frac{dy}{dx} = (1-x)^3$.

The curve passes through the point $(-1, 1)$.

Find the equation of the curve.

3

(c) If $\cos \beta = \frac{2}{5}$ and $\sin \beta < 0$, find the exact value of $\tan \beta$.

2

(d) Find the exact value of $\operatorname{cosec}\left(\frac{5\pi}{3}\right)$.

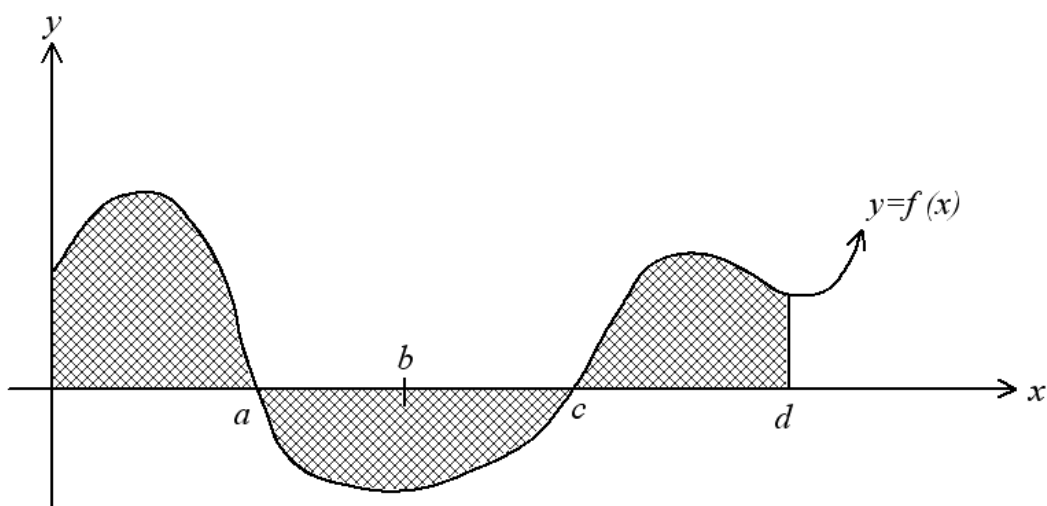
2

(e) Express $\frac{2}{\sqrt{5}-1} - \frac{3}{\sqrt{5}+1}$ in its simplest form.

2**End of Question 6**

Question 7 (12 marks) **[START A NEW PAGE]**

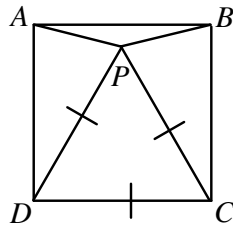
- (a) A town's population was recorded at the start of 2004. The population, P , t years later is given by the exponential equation $P = 120000e^{-0.05t}$.
- (i) What was the initial population of the town at the start of 2004? **1**
 - (ii) Find the time in years it will take the initial population to halve. **2**
 - (iii) At what rate is the population changing at the start of 2010? **2**
- (b) (i) Show the locus of a point $P(x, y)$ which moves so that $PA = 2PB$, where A is the point $(-2, 4)$ and B is $(4, 1)$ is:
- $$x^2 + y^2 - 12x + 16 = 0$$
- 3**
- (ii) Find the centre and radius of the circle defined in (i). **2**
- (c) Write an expression for the shaded area shown in the diagram below. **2**



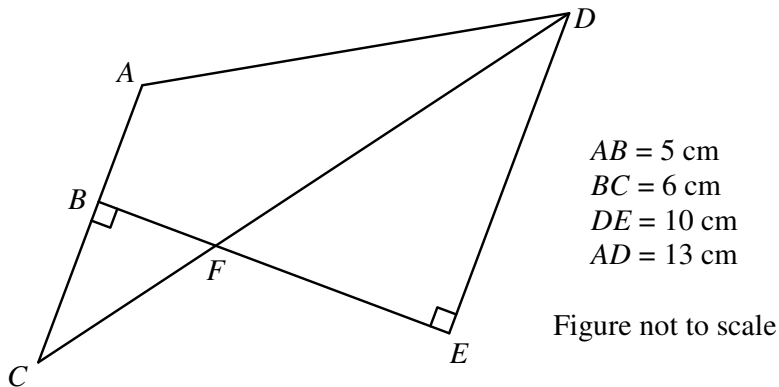
End of Question 7

Question 8 (12 marks) [START A NEW PAGE]

- (a) P is a point inside a square ABCD such that triangle PDC is equilateral. Prove that:



- (i) $\triangle APD \equiv \triangle BPC$. 3
- (ii) $\triangle APB$ is isosceles. 1
- (b) Copy the following diagram onto your answer sheet.



- (i) Prove $\triangle BFC \parallel \triangle EFD$. 2
- (ii) Find the length of DF. 3
- (c) Sketch the parabola $(y-1)^2 = -8(x+2)$, clearly showing the:
- (i) coordinates of the vertex 1
- (ii) coordinates of the focus 1
- (iii) equation of the directrix 1

End of Question 8

Question 9 (12 marks) **[START A NEW PAGE]**

Two particles A and B move along the x -axis, both starting when $t = 0$.

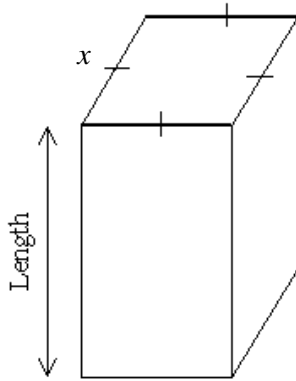
The displacement of particles A and B is given by $x = t + 12 - t^2$ and $x = t^2 - 4t$ respectively. In both cases x is the displacement of the particle from O in metres and t is measured in seconds.

- | | | |
|-------|--|---|
| (i) | Find when and where particle A is stationary. | 2 |
| (ii) | On the same diagram, sketch each particle's displacement graph, showing all intercepts. | 3 |
| (iii) | Show that the distance, D , between the two particles during $0 \leq t \leq 4$, is given by $D = 5t + 12 - 2t^2$. | 1 |
| (iv) | During the first 4 seconds, when are the particles furthest apart? | 2 |
| (v) | Find the time when both particles have the same velocity. | 2 |
| (vi) | Make a statement about the accelerations of the particles, being sure to justify your statement with the aid of mathematical evidence. | 2 |

End of Question 9

Question 10 (12 marks) **[START A NEW PAGE]**

- (a) A block of wood is in the shape of a square-based prism.
The **sum** of the length of the block and the perimeter of the base is 12 cm.



- (i) Show that $V = 12x^2 - 4x^3$, where V is the volume of the block. **2**
- (ii) What is the volume of the largest block? **3**
- (b) (i) Greg borrows \$400000 in order to buy an apartment. The interest rate is 6% p.a. reducible and the loan is to be repaid in equal monthly repayments of \$ M over 25 years with the interest calculated monthly. Let \$ A_n be the amount owing after the n th repayment.
- (α) Write down expressions for \$ A_1 and \$ A_2 , the amounts owing after the first and second repayments have been made respectively. **1**
- (β) Show that the amount of each monthly repayment is \$2577.21 (correct to the nearest cent). **2**
- (ii) After 5 years (i.e. 60 repayments) the interest rate rises to 9% p.a. Find the new monthly repayment \$ N , correct to the nearest cent. (Assume that the period of the loan is still 25 years). **3**
- (iii) How much extra does Greg repay over the life of the loan as a result of the 3% p.a. interest rate rise? **1**

End of Question 10**End of Paper**

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \quad \text{if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

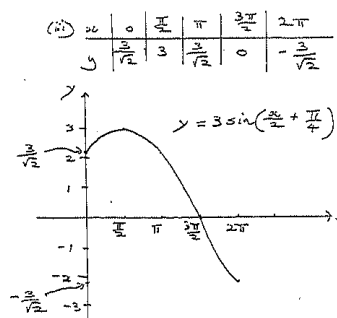
$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

④ (a) (i) Amp. = 3 units

(ii) Period = $\frac{2\pi}{\frac{1}{2}} = 4\pi$



(b) $4 \sin x = 3 \operatorname{cosec} x$

$4 \sin x = \frac{3}{\sin x}$

$\therefore \sin^2 x = \frac{3}{4}$

$\sin x = \pm \frac{\sqrt{3}}{2}$

Basic angle = 60°

$x = 60^\circ, 180^\circ - 60^\circ, 180^\circ + 60^\circ, 360^\circ - 60^\circ$

$x = 60^\circ, 120^\circ, 240^\circ, 300^\circ$

(c) Prove $\frac{\cos x}{1 + \sin x} = \sec x (1 - \sin x)$

LHS = $\frac{\cos x}{1 + \sin x} \times \frac{1 - \sin x}{1 - \sin x}$

= $\frac{\cos x (1 - \sin x)}{1 - \sin^2 x}$

= $\frac{\cos x (1 - \sin x)}{\cos^2 x}$

= $\frac{1 - \sin x}{\cos x}$

= $\sec x (1 - \sin x)$

= RHS

(d)

$(-6) + (-2) + 2 + \dots$

$T_2 - T_1 = 4 = T_3 - T_2 \Rightarrow$ AP with $d = 4, a = -6$

$T_{32} = a + 31d$
 $= -6 + 31 \times 4$
 $= 118$

(e) $108 + 36 + 12$

$r = \frac{36}{108} = \frac{12}{36} = \frac{1}{3}$

$S_\infty = \frac{108}{1 - \frac{1}{3}} = \frac{108}{\frac{2}{3}} = 162$

⑤ (a)

$\int \frac{dx}{\sqrt{x^2}} = \int \frac{dx}{x^{\frac{1}{2}}}$

= $\int x^{-\frac{1}{2}} dx$

= $\frac{x^{\frac{1}{2}}}{\frac{1}{2}} + C$

= $2\sqrt{x} + C$

(b) $\int_2^{\sqrt{7}} \frac{2x}{x^2-3} dx = \frac{1}{2} \int_2^{\sqrt{7}} \frac{2x}{x^2-3} dx$

= $\frac{1}{2} \left[\ln(x^2-3) \right]_2^{\sqrt{7}}$

= $\frac{1}{2} \{ (\ln 4) - (\ln 1) \}$

= $\frac{1}{2} \ln 4 = \ln 2$

= $\ln 2$

x	0	1	2
f(x)	1	2.718	7.389

(i) $\int_0^2 x^2 dx = \frac{1}{3} \{ y_1 + y_2 + y_3 \}$

= $\frac{1}{3} \{ 1 + 2.718 + 7.389 \}$

= 6.420333333

(d) $V = \pi \int_1^5 (-2x)^2 dx$

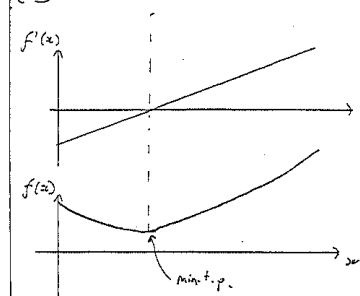
= $\pi \int_1^5 x^2 dx$

= $\pi \left[\frac{x^3}{3} \right]_1^5$

= $\frac{\pi}{3} \{ 125 - 1 \}$

= $\frac{496\pi}{3}$

(e)



⑥

$$(a) y = \log_e \left(\frac{x}{2} \right)$$

$$\therefore e^y = \frac{x}{2}$$

$$\therefore x = 2e^y$$

$$\text{Req'd Area} = 2 \int_0^{\ln 4} e^y dy$$

$$= 2 \left[e^y \right]_0^{\ln 4}$$

$$= 2 \{ e^{\ln 4} - e^0 \}$$

$$= 2 \{ 4 - 1 \}$$

$$= \underline{6 \text{ u}^2}$$

$$(b) \frac{dy}{dx} = (1-x)^3$$

$$y = \frac{(1-x)^4}{(-1) \times 4} + C$$

sub in $(-1, 1)$:

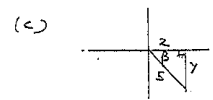
$$1 = \frac{(1+1)^4}{-4} + C$$

$$1 = -4 + C$$

$$C = 5$$

$$\therefore y = \underline{-\frac{1}{4}(1-x)^4 + 5}$$

(c)



$$y^2 + 2^2 = 5^2$$

$$y^2 = 21$$

$$y = -\sqrt{21} \quad (y < 0)$$

$$\therefore \tan \beta = \underline{\underline{-\frac{\sqrt{21}}{2}}}$$

$$(d) \operatorname{cosec} \left(\frac{5\pi}{3} \right)$$

$$= \frac{1}{\sin \left(\frac{5\pi}{3} \right)}$$

$$= \frac{1}{-\sin \frac{\pi}{3}}$$

$$= \frac{1}{-\frac{\sqrt{3}}{2}}$$

$$= \underline{\underline{-\frac{2}{\sqrt{3}}}}$$

$$(e) \frac{2}{\sqrt{5}-1} - \frac{3}{\sqrt{5}+1}$$

$$= \frac{2(\sqrt{5}+1) - 3(\sqrt{5}-1)}{(\sqrt{5}-1)(\sqrt{5}+1)}$$

$$= \frac{2\sqrt{5}+2-3\sqrt{5}+3}{5-1}$$

$$= \underline{\underline{\frac{5-\sqrt{5}}{4}}}$$

⑦

$$(a) (i) \text{ when } t=0, P = 120000$$

$$(ii) \text{ when } P = 60000, \quad 60000 = 120000 e^{-0.05t}$$

$$\therefore e^{-0.05t} = \frac{1}{2}$$

$$\therefore -0.05t = \ln \left(\frac{1}{2} \right)$$

$$\therefore t = \frac{\ln \left(\frac{1}{2} \right)}{-0.05}$$

$$= 13.86294361$$

$$= \underline{13.9 \text{ years (1 d.p.)}}$$

$$(iii) \frac{dP}{dt} = -0.05 \times 120000 e^{-0.05t}$$

$$= -6000 e^{-0.05t}$$

$$\text{when } t=6, \frac{dP}{dt} = -6000 e^{-0.05 \times 6}$$

$$= -6000 e^{-0.3}$$

$$= -4444.909324$$

$$\text{Req'd rate} = \underline{\underline{-4445 \text{ p.a.}}}$$

$$(b) (i) PA = 2PB$$

$$\Rightarrow \sqrt{(x+2)^2 + (y-4)^2} = 2\sqrt{(x-4)^2 + (y-1)^2}$$

$$\therefore (x+2)^2 + (y-4)^2 = 4 \{ (x-4)^2 + (y-1)^2 \}$$

$$\therefore x^2 + 4x + 4 + y^2 - 8y + 16$$

$$= 4 \{ x^2 - 8x + 16 + y^2 - 2y + 1 \}$$

$$\therefore x^2 + y^2 + 4x - 8y + 20$$

$$= 4x^2 + 4y^2 - 32x - 8y + 68$$

$$\therefore 0 = 3x^2 + 3y^2 - 36x + 48 = 0$$

$$\therefore x^2 + y^2 - 12x + 16 = 0$$

$$(ii) (x^2 - 12x + 36) + y^2 = 20$$

$$(x-6)^2 + y^2 = 20$$

$$\Rightarrow \text{centre is } (6, 0)$$

$$\Leftarrow \text{radius} = \underline{\underline{\sqrt{20} \text{ units}}}$$

(c)

Shaded area

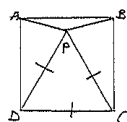
$$= \int_0^2 f(x) dx + \left| \int_2^c f(x) dx \right| + \int_c^d f(x) dx$$

OR

Shaded area

$$= \int_0^2 f(x) dx - \int_2^c f(x) dx + \int_c^d f(x) dx$$

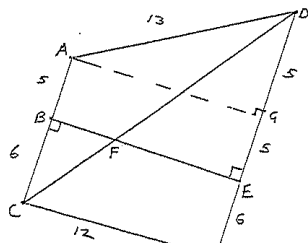
8 (2)



- (i) $\angle ADC = \angle BCD = 90^\circ$ ($\angle$'s of square)
 $\angle PDC = \angle PCD = 60^\circ$ ($\angle$'s of equilateral Δ)
 $\therefore \angle ADP = 30^\circ = \angle BCP$ (difference of equal amounts)
 $AD = BC$ (sides of a square)
 $PD = PC$ (sides of equilateral Δ)
 $\therefore \Delta ADP \cong \Delta BCP$ (SAS)

- (ii) $AP = BP$ (corresponding sides of congruent Δ 's)
 $\therefore \Delta APB$ is isosceles.

(b)



- (i) $\angle CBF = 90^\circ = \angle DEF$ (given)
 $\angle BFC = \angle DFE$ (vertically opposite $\angle$'s)
 $\therefore \Delta BFC \cong \Delta EFD$ (equilateral).

- (ii) $AG = \sqrt{13^2 - 5^2}$
 $= 12$
 $= BE = CH$
 $CD = \sqrt{12^2 + 16^2}$
 $= 20$

Let $DF = x \Rightarrow CF = 20 - x$

$\therefore \frac{x}{10} = \frac{20-x}{6}$ { corresponding sides of sim. Δ 's are in same ratio }

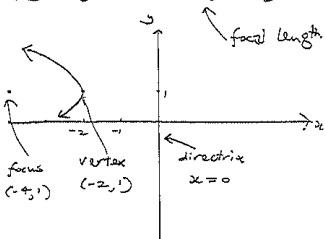
$\therefore 6x = 200 - 10x$

$16x = 200$

$x = 12.5$

$\therefore DF = 12.5 \text{ cm}$

(c) $(y-1)^2 = -4(x+2)$



9

(i) $x_A = t + 12 - t^2$

$\frac{dx_A}{dt} = 1 - 2t$

When $\frac{dx_A}{dt} = 0$, $1 - 2t = 0$

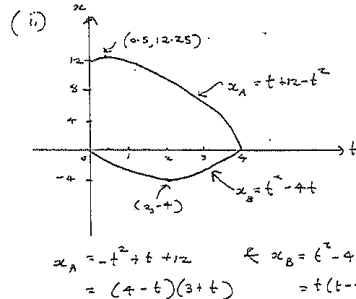
$\therefore t = \frac{1}{2}$

When $t = \frac{1}{2}$, $x_A = \frac{1}{2} + 12 - \frac{1}{4}$

$= 12.25 \text{ m}$

$\therefore A$ is dist. at $x = 12.25 \text{ m}$

after 0.5 s.



$x_A = -t^2 + t + 12 \quad \& \quad x_B = t^2 - 4t$
 $= (4-t)(3+t) \quad = t(t-4)$

(iii) for $0 \leq t \leq 4$,

$D = x_A - x_B$
 $= t + 12 - t^2 - (t^2 - 4t)$
 $= t + 12 - t^2 - t^2 + 4t$
 $= 5t + 12 - 2t^2$

(iv) $\frac{dD}{dt} = 5 - 4t$

$\& \quad \frac{d^2D}{dt^2} = -4 \Rightarrow$

When $\frac{dD}{dt} = 0$, $4t = 5$
 $\therefore t = \frac{5}{4}$

given distance function is concave down for all t , D is a maximum at $t = \frac{5}{4}$.
Hence A & B are furthest apart after 1.25 s

(v)

$\ddot{x}_A = 1 - 2t \quad \& \quad \ddot{x}_B = 2t - 4$

$\ddot{x}_A = \ddot{x}_B \Rightarrow 1 - 2t = 2t - 4$

$\therefore 4t = 5$

$\therefore t = 1.25 \text{ s}$

(vi) $\ddot{x}_A = -2 \text{ ms}^{-2}$

$\ddot{x}_B = 2 \text{ ms}^{-2}$

Both accelerations are constant and the same magnitude; however A 's is positive & B 's is negative.

(10)

(a)(i)

$$l + 4x = 12$$

$$\therefore l = 12 - 4x$$

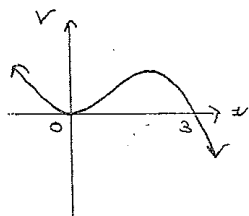
$$V = x \times x \times (12 - 4x)$$

$$= x^2(12 - 4x)$$

$$= 12x^2 - 4x^3$$

(ii)

$$V = 4x^2(3 - x)$$



$$\Rightarrow 0 < x < 3$$

$$\frac{dV}{dx} = 24x - 12x^2$$

$$\frac{d^2V}{dx^2} = 24 - 24x$$

$$\frac{dV}{dx} = 0 \Rightarrow 12x(2 - x)$$

$$\Rightarrow x = 0 \text{ or } x = 2$$

X

$$\text{When } x = 2, \frac{d^2V}{dx^2} = 24 - 48$$

$$= -24$$

 $< 0 \Rightarrow \text{max.}$

$$\text{When } x = 2, V = 48 - 32$$

$$= 16$$

$$\therefore \text{Max. vol.} = 16 \text{ m}^3$$

(b)(i)(a)

$$A_1 = 400000 \times 1.005 - M$$

$$A_2 = 400000 \times 1.005^2 - 1.005M - M$$

$$(\beta) A_{300} = 400000 \times 1.005^{300} - M(1 + 1.005^1 + \dots + 1.005^{299})$$

$$\text{But } A_{300} = 0, \text{ so}$$

$$M = \frac{400000 \times 1.005^{300}}{(1.005^{300} - 1) \times 0.005}$$

$$= 2577.205606$$

$$= \$2577.21 \text{ (n.cent)}$$

$$(ii) A_{60} = 400000 \times 1.005^{60} - 2577.21(1 + 1.005^1 + \dots + 1.005^{59})$$

$$= 400000 \times 1.005^{60} - 2577.21 \times \frac{1.005^{60} - 1}{0.005}$$

$$= 359728.0407$$

$$= \$359728.04 \text{ (n.cent)}$$

$$A_{61} = 359728.04 \times 1.0075 - N$$

$$A_{62} = 359728.04 \times 1.0075^2 - 1.0075N - N$$

$$A_{300} = 359728.04 \times 1.0075^{240}$$

$$= N(1 + 1.0075 + \dots + 1.0075^{239})$$

$$\text{But } A_{300} = 0, \text{ so}$$

$$N = \frac{359728.04 \times 1.0075^{240}}{(1.0075^{240} - 1) \times 0.0075}$$

$$= 3236.566546$$

$$\therefore \text{Req'd amt} = \$3236.57 \text{ (n.cent)}$$

$$(iii) \text{ Extra paid} = \$ (3236.57 - 2577.21) \times 240$$

$$= \$158246.40$$